Thantham Khongsamret | Research Summary

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Abstract / scope

Predefined SPS-DAB and asymmetric CLLC bidirectional DC-DC converter implementations were compared in PSIM for a 400-V DC link, 250-450-V battery, 1-6-kW operating power, and G2V and V2G directions.
A total of 60 voltage-power-direction operating-point pairs were numerically qualified under the study's criteria.

Nominal modeled-efficiency results

CLLC led 31 of 60 paired operating points, mainly at light load.
SPS-DAB led 29 pairs, including 26 of 30 pairs at 4-6 kW.
These are conditional results under predefined controls, not universal topology rankings and not measurements on hardware.

Read full project page

https://t-khongsamret.github.io/research/p00167/
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This two-page research summary is not the six-page accepted manuscript.

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Study boundaries and interpretation

Numerical screening

Evidence categories: 12 supported, 44 warning, 4 additional validation.
Thirty-three of 60 nominal orderings are sensitive to differential
omitted-loss thresholds of 10 W or less.

Important limitations

- DAB uses explicit MOSFETs while CLLC has parameterized bridges.
- Local explicit-MOSFET checks do not bound all 60 operating pairs.
- Several passive and auxiliary losses are omitted from the model.
- ZVS/ZCS diagnostics are not experimental switching measurements.
- A hardware performance comparison has not been established.

Separate educational slide deck

Canva power, loss and PSIM notes include DAB 10-kW examples.
Those examples are distinct from the 1-6-kW paper data.
https://www.canva.com/d/5vcHAJwCWstG7k5

Status / citation

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